

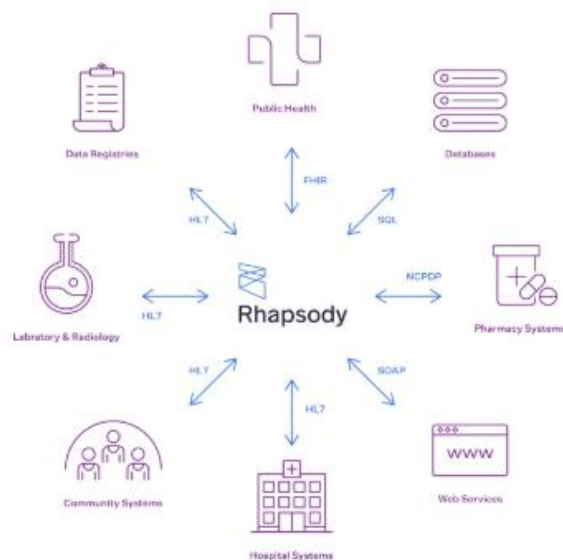
Rhapsody Associate Course Review

The *Rhapsody Associate Certification* course is intended for new and existing Rhapsody users wishing to improve their understanding of the underlying concepts of the application and to develop their skills in efficient route monitoring and configuration.

The course is a precursor to the *Rhapsody Professional Certification* course, which provides a good, practical understanding of the purpose of the Rhapsody Integration Engine and the range of problems its use is designed to solve.

Rhapsody Associate Certification Topics

1. What is a Message, and how is it processed by the Rhapsody engine?
2. What is Message Intelligence, and how can it be used to selectively route messages depending on their structure or content?
3. What are the Rhapsody Messaging Toolkit, and how do they help in the successful configuration of a Rhapsody installation?
4. What are the Best Practice principles that will help you build successful and efficient routes that meet the needs of your organization?
5. How is the Rhapsody Management Console used to monitor message processing, including the identification of potential bottlenecks and the management of messages that fail processing?



Certification Assessment

The course is divided into 10 modules of learning plus a final assessment.

The final assessment brings together an evaluation of the course learning with a final quiz and a practical exercise where a route is built according to the provided specifications.

REN-101: The Rhapsody Integration Engine

This module introduces the *Rhapsody Associate* course and provides background information on navigating through the modules, reviewing your progress and what is required before you can begin the Final Assessment. The concept of an Integration Engine is introduced, along with the implementation and communication models used by Rhapsody when it processes messages.

The four major Rhapsody components are introduced:

1. The Rhapsody Engine, which processes messages, has no user interface.
2. The Rhapsody IDE (Integrated Development Environment) along with its major parts:
 - General screen layout, including the Workspace, Route Editing Canvas, Filter Toolbox and the Output, Properties and Help panels.
3. The Monitoring Console, used to view the health of the Rhapsody engine and to search for and view the contents of messages from the archive.
4. The EDI Tools, including:
 - EDI Explorer, used to construct test messages and send them to the Rhapsody engine.
 - EDI Message Designer, used to construct files that 'tell' the Rhapsody engine of the structure of the messages it is processing.
 - Map designer, used to transform messages from one format to another.

REN-102: Working with Messages

This module introduces the concept of a Message, including its content and associated properties. The ways in which messages can be interpreted by the engine are described, along with character encoding, batched messages and the effect of the FIFO settings on message throughput.

Two exercises are included:

1. The first explores the ways in which HL7 and CSV messages are processed.
2. The second compares processing for single and batched messages, and investigates how the order in which messages are processed can be determined.

REN-103: Building a Simple Route

This module reviews the principles of simple route design, with an emphasis on implementing Rhapsody's best practice principles to ensure that the route is both robust and efficient. The use of input and output communication points to send and receive messages to and from Rhapsody is described, along with component check in and starting.

Features introduced in this module including the use of:

1. The IDE File Manager, used to maintain the list of message and mapper definitions loaded on to the engine.
2. Conditional Connectors to selectively route messages based on their structure and/or content.
3. Rhapsody Variables to configure solutions that are easy to maintain and migrate to other environments.

Rhapsody's Best Practice recommendations are an important part of this module. Topics covered include naming conventions for lockers, folders, route components and the names of definition files used in the configuration.

REN-104: Managing Your Configuration

Features introduced in this module include the use of:

1. Version Control to record a comment at the time of check-in, and to roll back to an earlier version of the configuration.
2. Saving and reloading configurations.
3. The IDE Management Screens, including the Group and User Manager, and the IDE Options.

REN-105: Communication Points

Concepts covered:

1. The Communication Points dialog.
2. Common communication points and their configuration.

Common communication points such as Directory, Sink, TCP Client, and TCP Server are introduced, and their configuration is explained. The configuration of communication points is managed by a set of properties accessible through the communication point properties dialog box. Many of these properties are common to other communication points.

REN-106: Filters

Concepts covered:

1. An overview of Filters, their construction and configuration.
2. The Filters dialog and the commonly used functionality.
3. Testing Filter functionality.
4. The Hold Queue.
5. Some commonly used Filters.

Introduces Filters and explains their ability to add business logic to routes and to control the path a message takes through the processing logic.

Introduces the Filter Toolbox grouping and commonly used filters. Explains Message processing flow in relation to Input, Output, Connectors, No-Match and Error.

Introduces the configuration of filters along with the concept of testing filter functionality.

Some other common filters are explained, such as No-Operation, HL7 Acknowledgement Generation, and Property Population filters.

REN-107: The Management Console

Concepts covered:

1. The Management Console.
2. Monitoring and Managing the Rhapsody engine.
3. Using the Management Console for error tracking.
4. The Management Console's search function.

Explores the Rhapsody Management Console in more detail and includes indicators of the Rhapsody engine health. Examines Message throughput, available memory and disk space along with processing, Error and Hold queues.

Explores the ability to manage the state of components individually or in bulk. Examines the detailed Message View, where the Message Event tree, Message Properties and Message Body can be evaluated.

The console's search function is explored and details the great flexibility in the ability to search for messages by Input, Output, and Route and including text and customizable searches.

REN-108: Route Building

Concepts covered:

1. Configuration Best Practices.

2. Finding components.
3. Route Interconnectivity.
4. Testing the functionality of the route and its components.

Previously covered concepts are brought together in order to construct a complex solution involving two interconnected routes.

REN-109: Interpreting Messages

Concepts covered:

1. A further investigation of the Rhapsody Messaging Toolkit and their functionality.
2. An understanding of fundamental message handling using the Rhapsody Messaging Toolkit.
3. Building and managing a message definition from the Library.
4. Building a Definition for a CSV Message.

The focus is on handling and processing of intelligent messages. The Rhapsody Messaging Toolkit and libraries are explored in more detail, showing how they provide Rhapsody with the ability to handle messages intelligently rather than handling them as atomic objects.

Explores creation of message definitions using the library. HL7, CSV and XML definitions are used to demonstrate how they affect message handling along the message pipeline.

REN-110: Message Transformation

Concepts covered:

1. The Message Modifier filter.
2. The Rhapsody Messaging Toolkit Mapper and its functionality.
3. The use of the Mapper to create, compile and test definition files and the transformation of messages between different structures.

One of the primary tasks often required of the Rhapsody engine is the transformation of messages from one format to another. This may arise due to the characteristics of upstream and downstream environments, such as where they support different HL7 levels or where one environment produces XML messages while the intended recipient requires HL7 messaging.